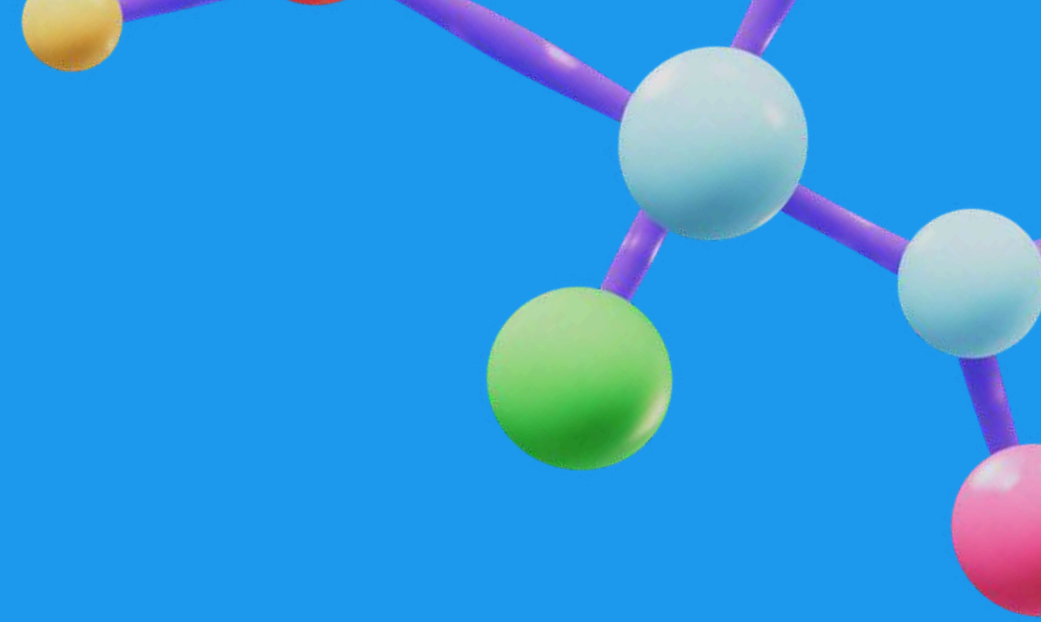
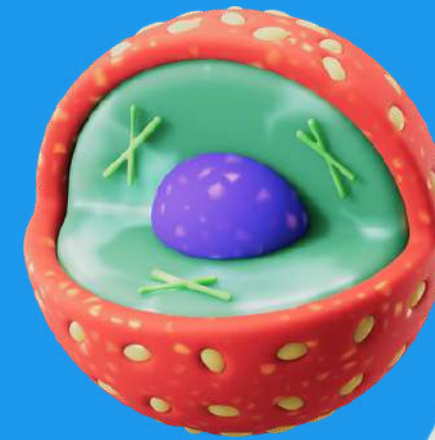
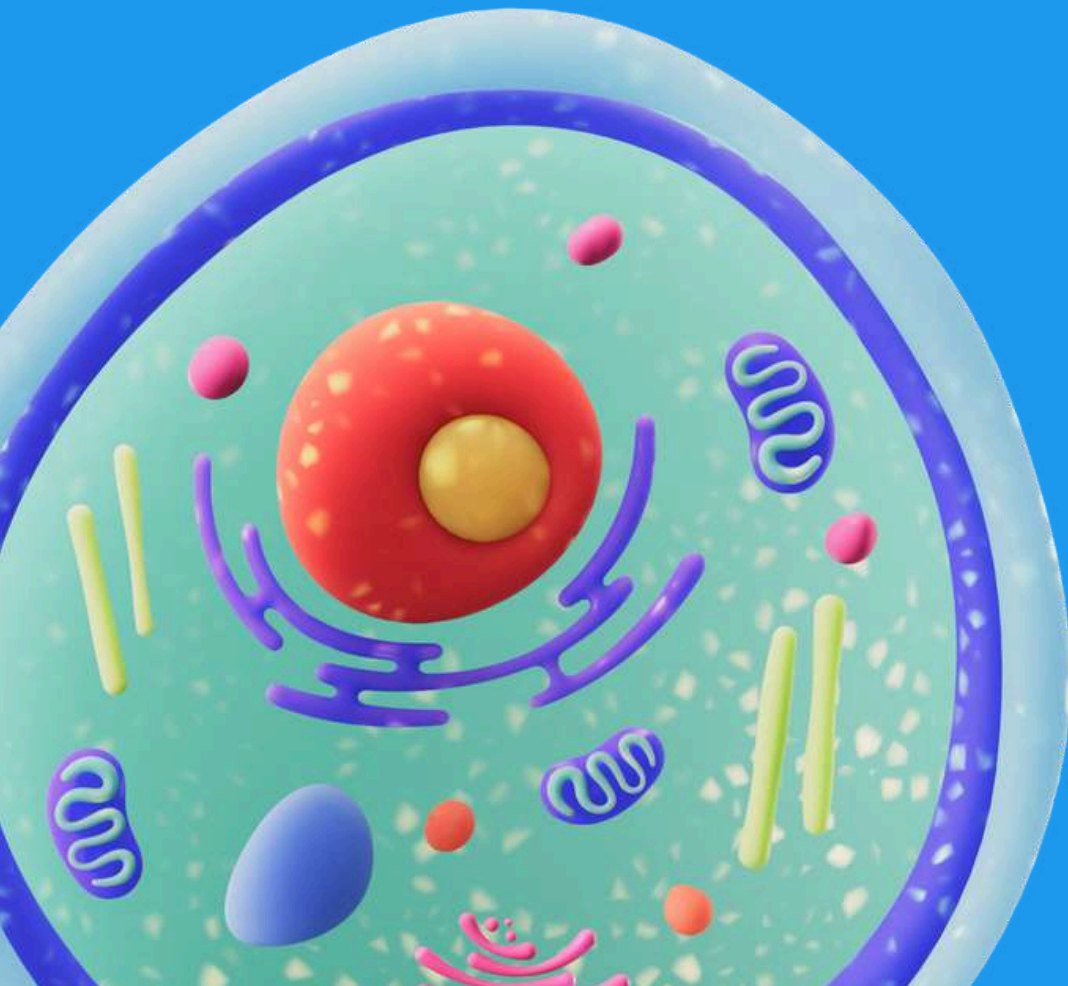


Cell

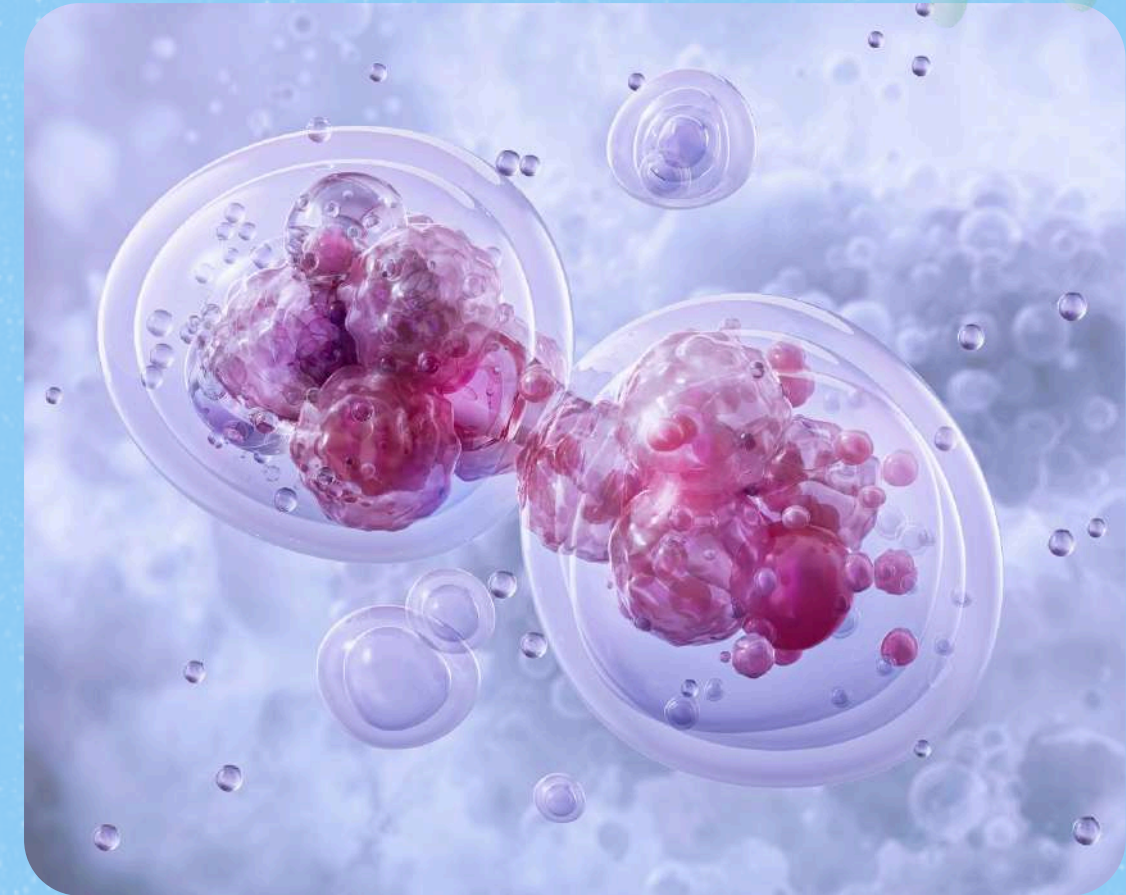
The basic structural and functional unit of life



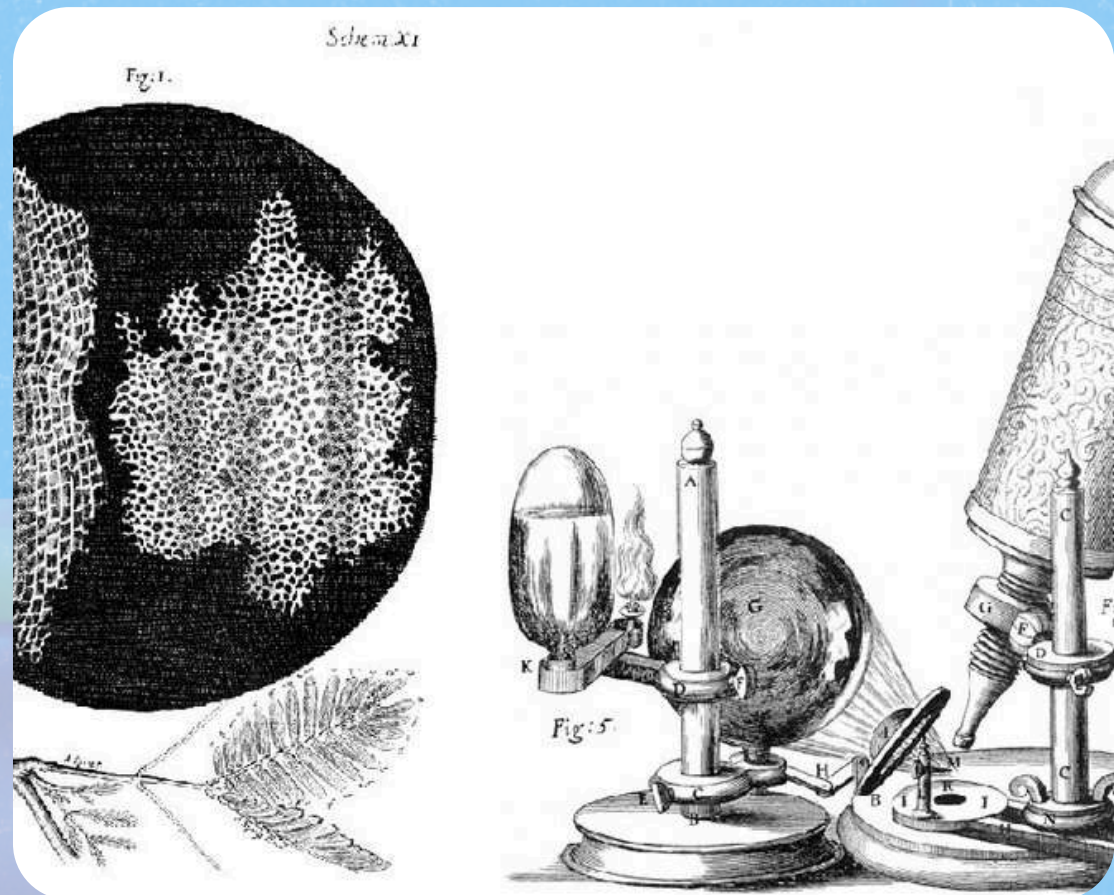
The CELL

What is a cell?

- Basic structural and functional unit of life.
- All living organisms are composed of cells and their products.
- Cell-callus-tissue-organ system-life
- Responsible for:
 - Nutrition
 - Respiration
 - Growth
 - Excretion
 - Reproduction
 - Response to stimuli
- Cells vary greatly in shape, size and function, but share fundamental characteristics.



Discovery of the cell



- The discovery of cells became possible after the development of the microscope.
- In 1665, English scientist Robert Hooke examined a dead cork cell (sclerenchyma) from bark of Spanish oak (*Quercus suber*) using a compound microscope. He saw a honeycomb-like hexagonal structure and observed the cell wall in the cell.

Discovery of the cell



**A. van
Leeuwenhoek**

Observed living
microscopic
organisms

1670



Robert Brown

Described the
nucleus

1831



M. Schleiden

Plants are made
of cells

1838



T. Schwann

Animals are
made of cells

1839



R. Virchow

New cells arise
from pre-
existing cells

1855

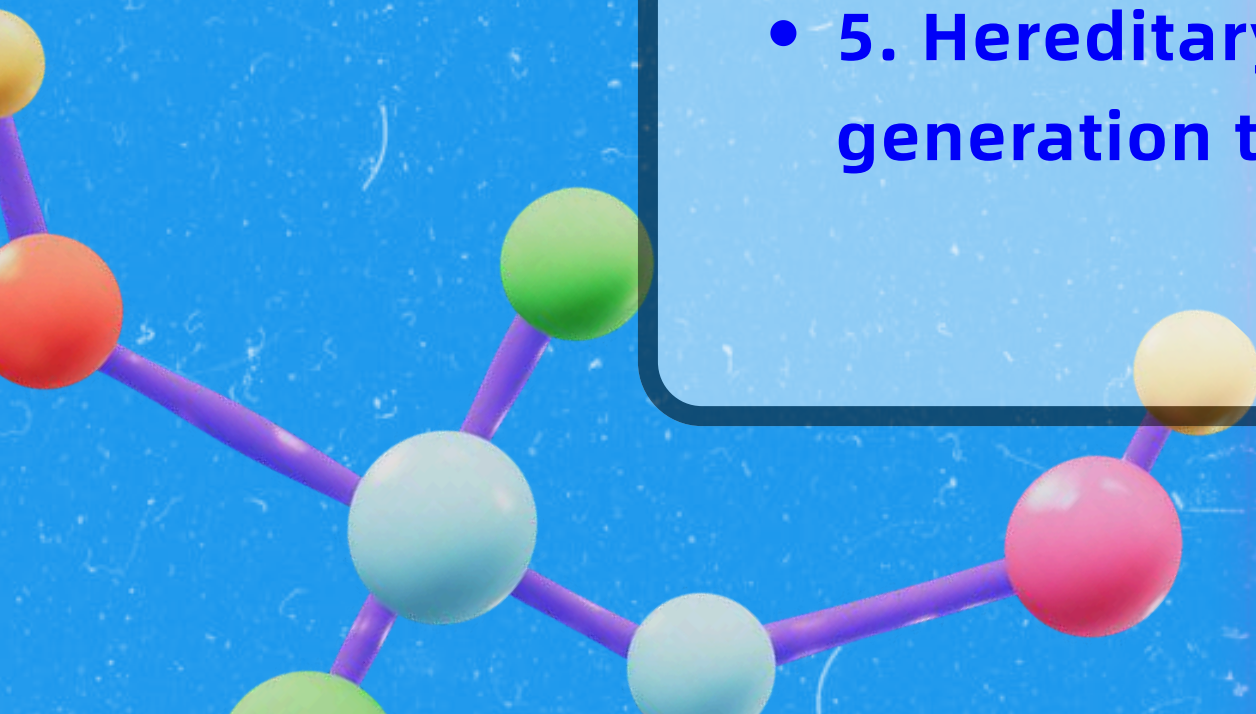
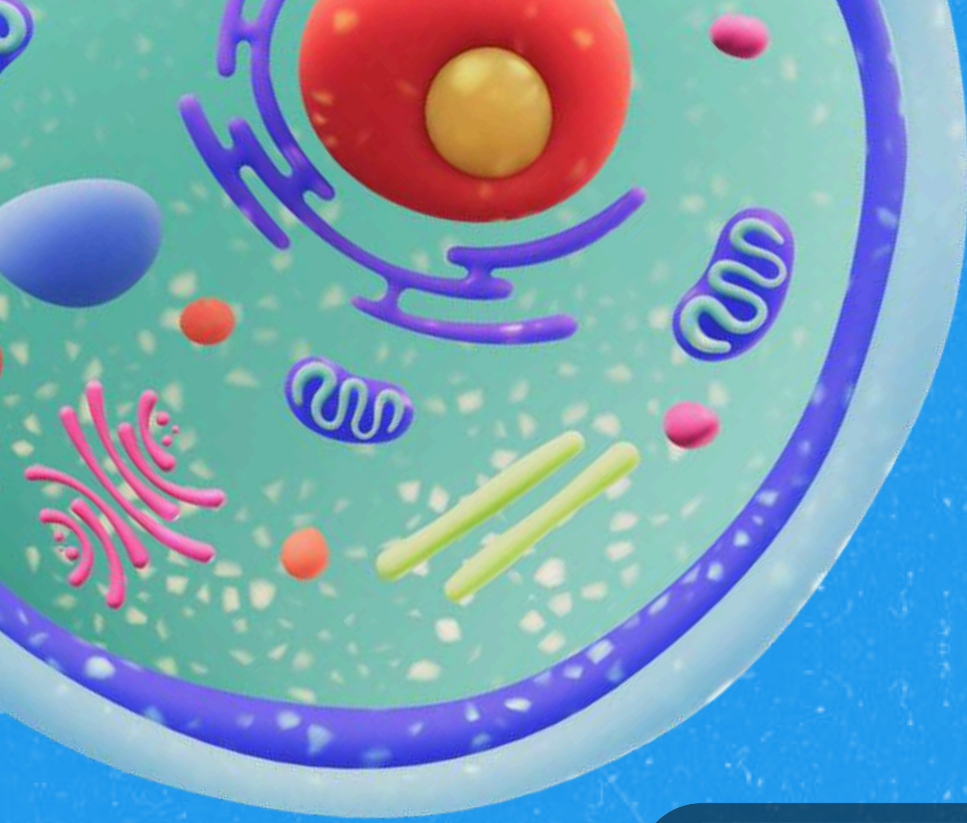
Cell Theory

Cell Theory

The cell theory was proposed by:

- M. J. Schleiden — Botanist, 1838
- Theodor Schwann — Zoologist, 1839

- 1. All living organisms are made up of cells and their products.
- 2. The cell is the basic structural and functional unit of life.
- 3. All cells arise from pre-existing cells.
- 4. Cells contain protoplasm with a nucleus and various organelles.
- 5. Hereditary information is passed from generation to generation through living cells.



Exceptions, Drawbacks & Limitations of Cell Theory

1. Viruses — Acellular

Viruses are acellular and lack a true cellular organization.

They are generally considered non-living outside a host cell and can reproduce only inside living cells.

2. Coenocytic / Multinucleate Organisms

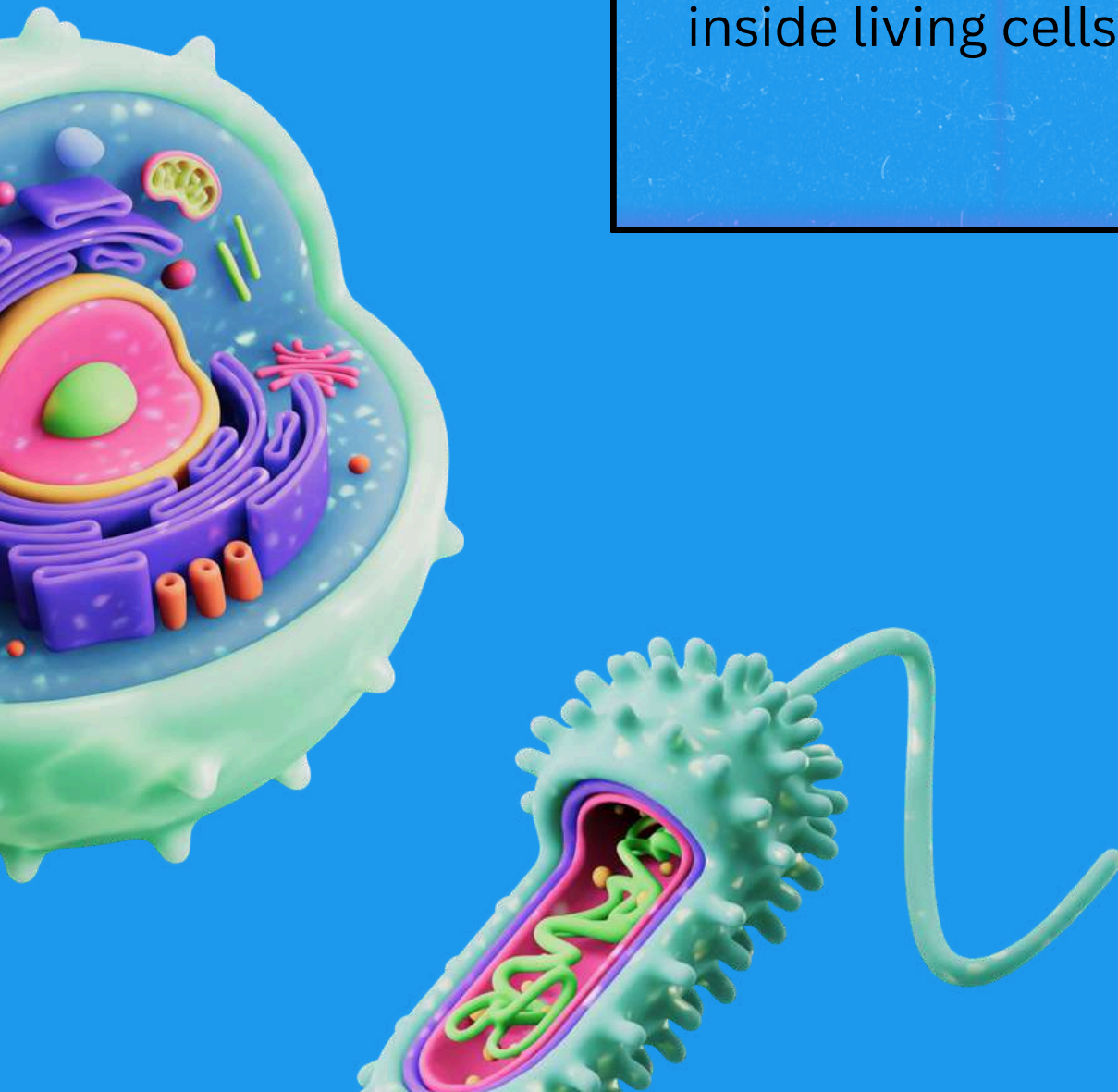
Some organisms contain many nuclei within a continuous cytoplasm, without distinct cell boundaries.

Examples include certain fungi and algae.

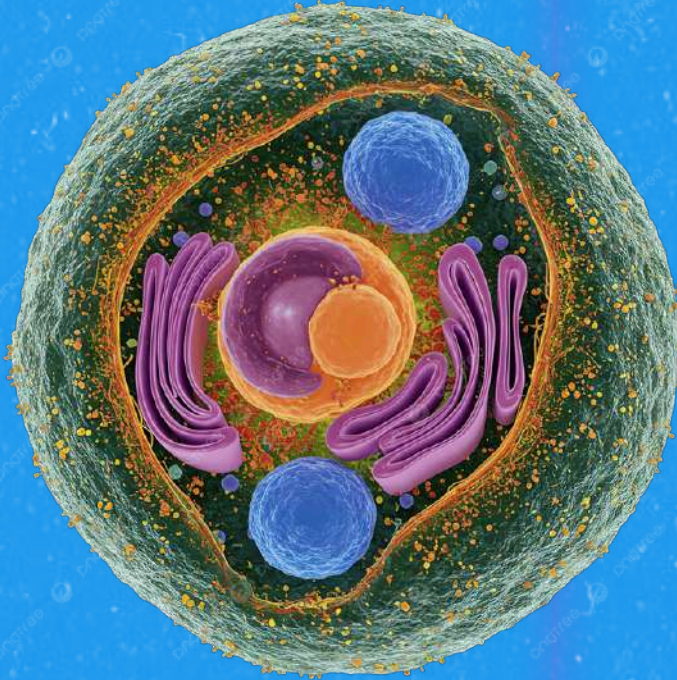
3. Limits of the Classical Cell Theory

Modern discoveries have shown that the original cell theory does not explain every biological structure perfectly.

Cell theory has therefore been expanded and modified with advances in biology.



Prokaryotic & Eukaryotic cells



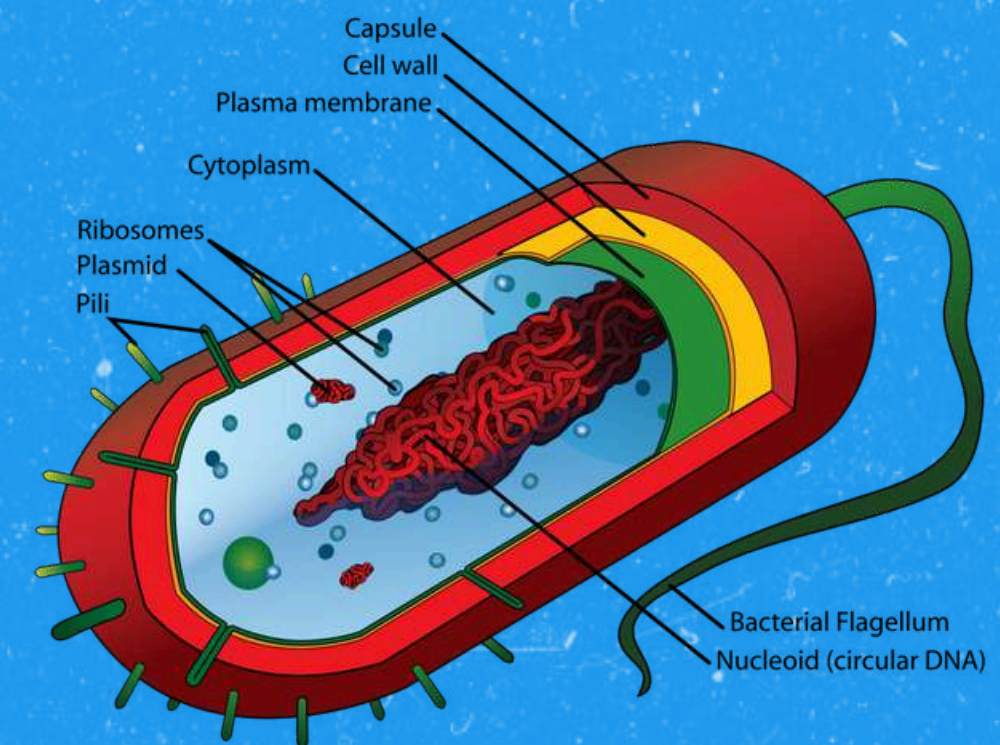
EUKARYOTIC

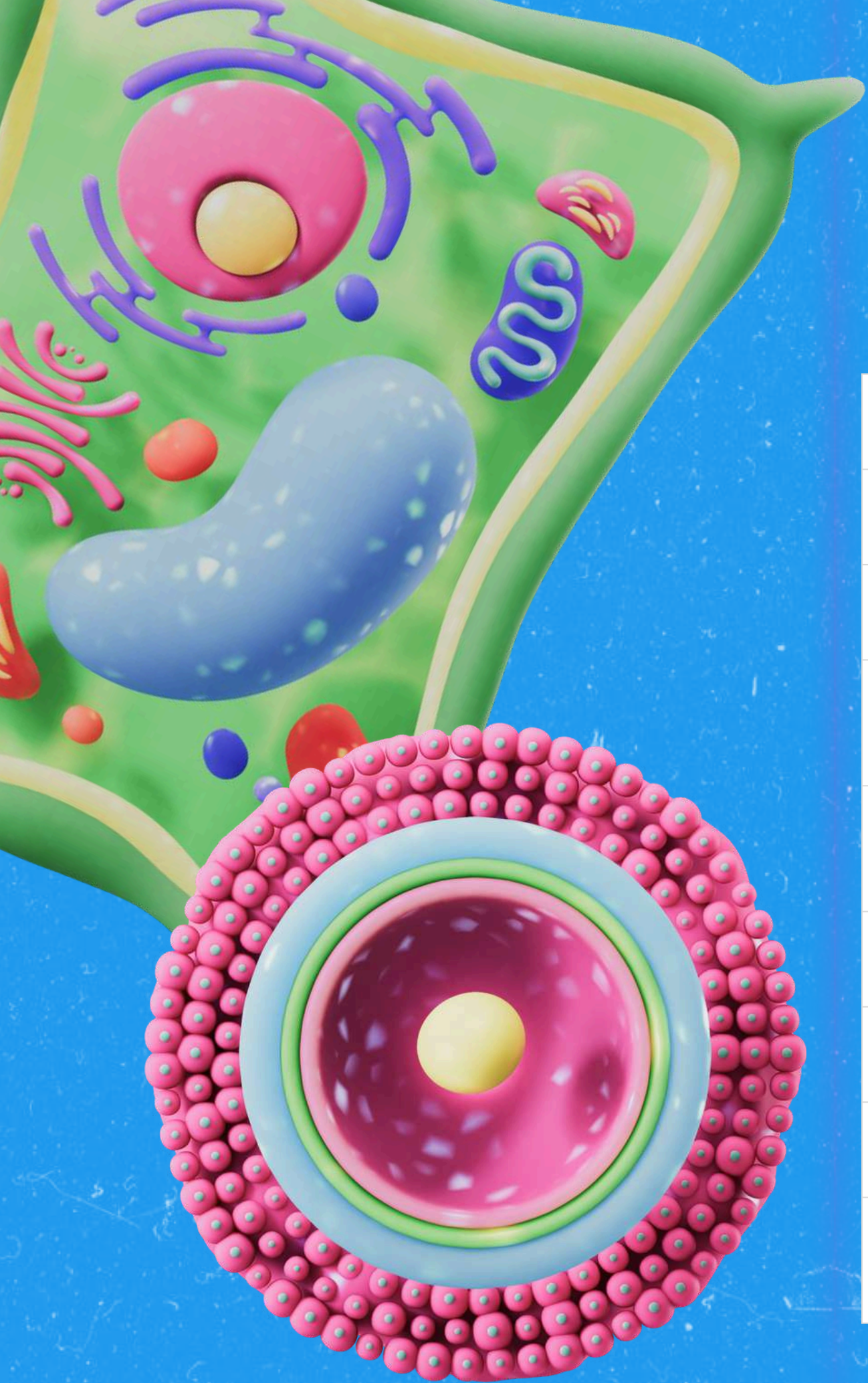
Simple & Primitive Cellular Organization

- No true nucleus — DNA lies in the nucleoid region.
- Nucleoid lacks a nuclear membrane, nucleoplasm and nucleolus.
- Generally small and structurally simple.
- Ribosomes are present for protein synthesis.
- Cell wall is present in many prokaryotes.
- Genetic material is generally circular DNA.
- Examples: Bacteria and Archaea.

PROKARYOTIC

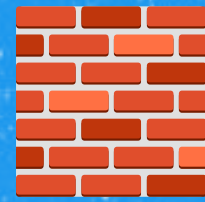
- Possesses a true, well-developed nucleus.
- Nucleus is surrounded by a nuclear membrane.
- Contains membrane-bound organelles.
- Generally larger and more complex than prokaryotic cells.
- Contains organelles
- Genetic material is organized into linear chromosomes.
- Found in plants, animals, fungi and protists.





Differences

Feature	 Prokaryotic	 Eukaryotic
Nucleus	No true nucleus	True nucleus
Nuclear membrane	Absent	Present
DNA	Usually circular	Usually linear chromosomes
Membrane-bound organelles	Absent	Present
Size	Generally smaller	Generally larger
Organization	Simple	Complex
Ribosomes	Present	Present
Examples	Bacteria, Archaea	Plants, animals, fungi, protists



CELL WALL: NATURE & LAYERS

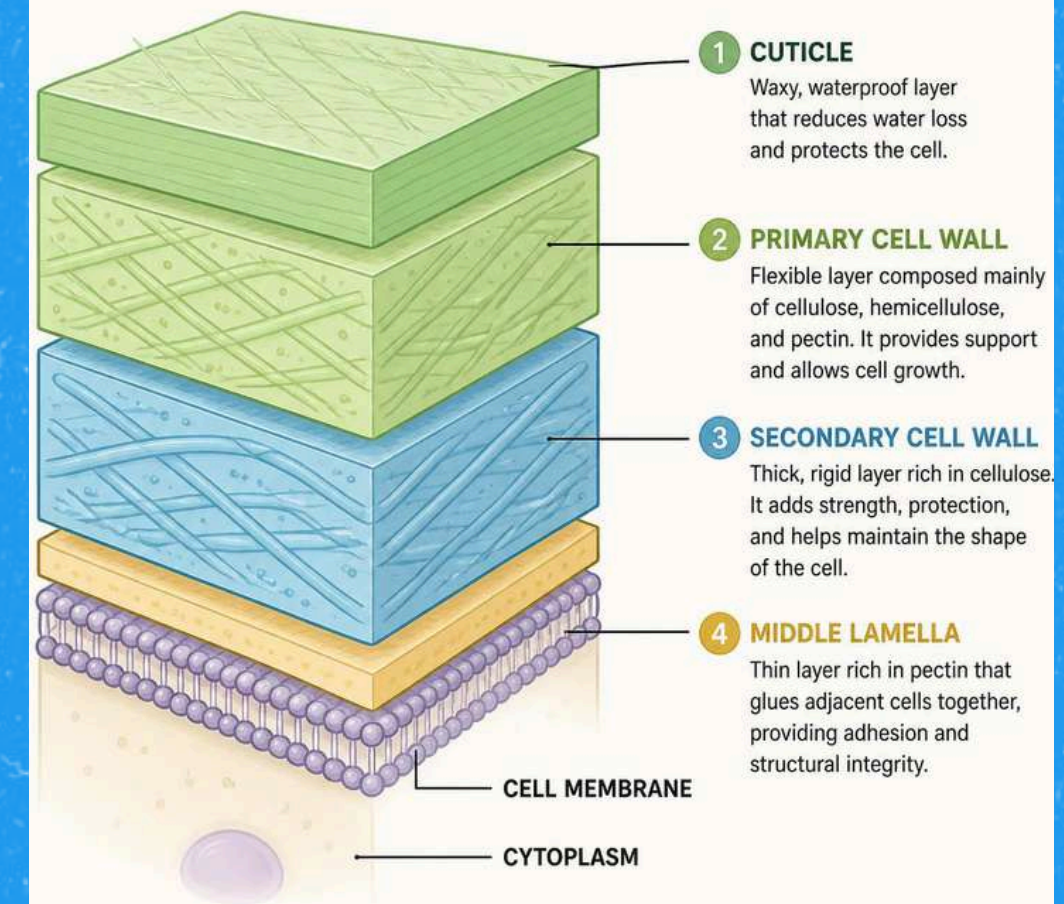
Nature

- Non-living, rigid and relatively inelastic layer.
- Present outside the plasma membrane in plant cells.
- Generally permeable to water and small substances.
- Provides shape, support and protection.

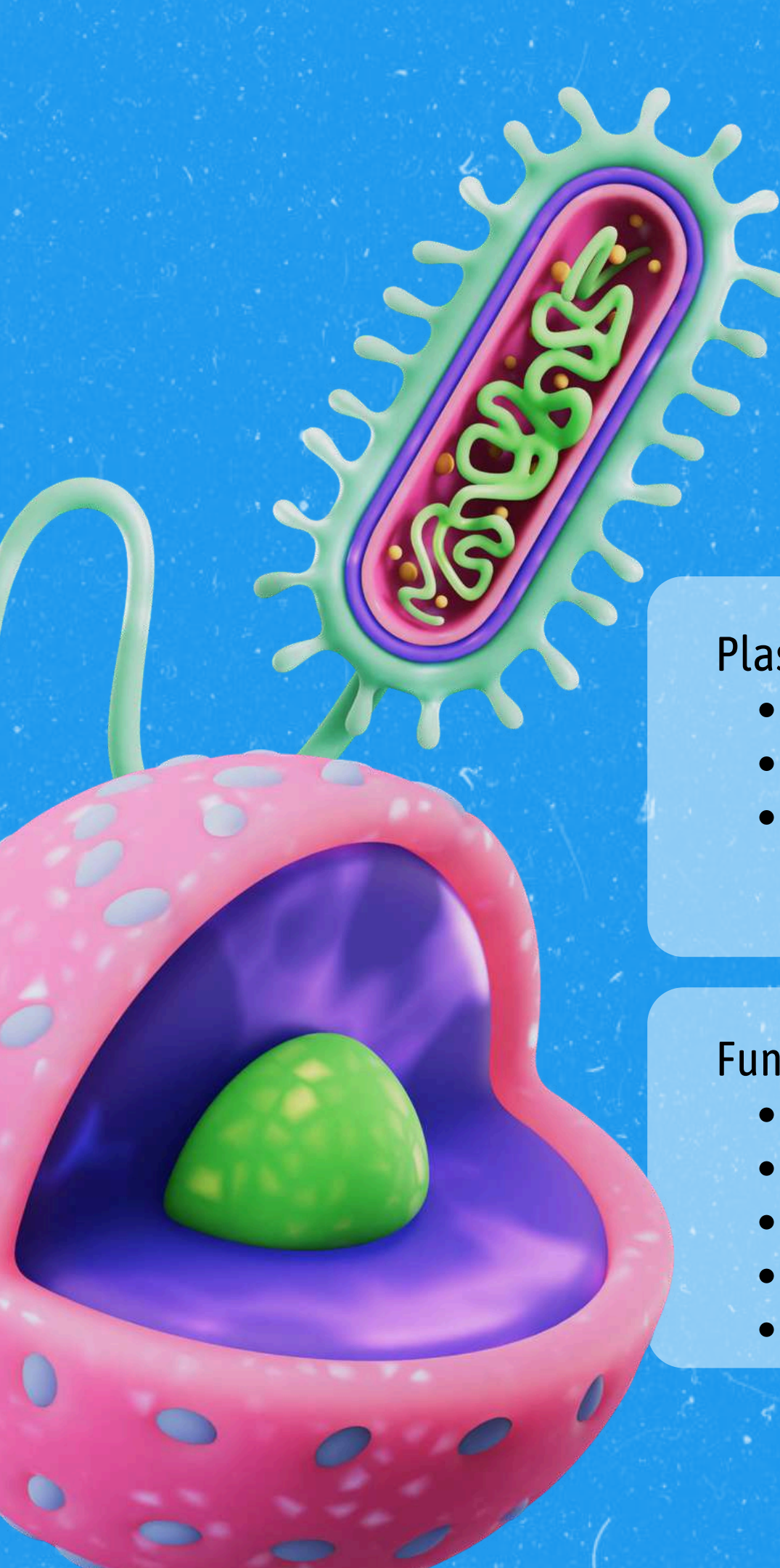
Layers

- Middle Lamella — cements adjacent cells; rich in pectates.
- Primary Wall — thin, flexible; cellulose + hemicellulose + pectin.
- Secondary Wall — thick, rigid; cellulose + lignin.
- Tertiary Wall — specialized innermost layer in some cells.

PLANT CELL WALL



Functions of Cell Wall



Plasmodesmata

- Tiny cytoplasmic channels connecting neighbouring plant cells.
- Pass through the cell wall.
- Enable intercellular transport and communication.

Functions of Cell Wall

- Provides mechanical support.
- Protects against mechanical damage.
- Maintains cell shape and rigidity.
- Helps withstand turgor pressure.
- Plasmodesmata facilitate cell-to-cell communication.

Cell Membrane

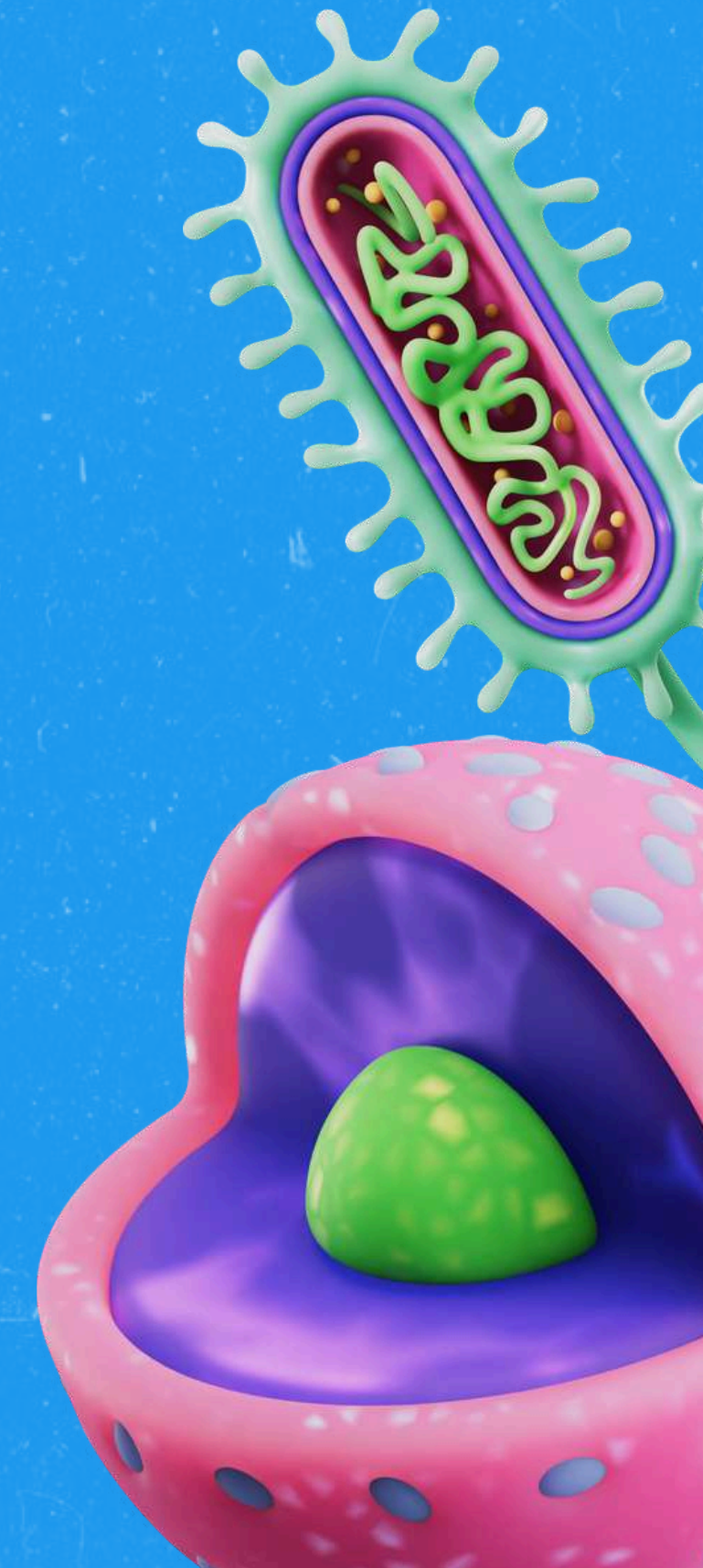
-A thin, flexible barrier that surrounds every living cell.
Their models are as follows:

A. Sandwich Model — Danielli & Davson (1935)

- The **Sandwich Model** was proposed by **Danielli and Davson in 1935**.
- According to this model, the cell membrane has **three main layers**: a **lipid bilayer** in the middle and **protein layers** on both sides.
- The lipid bilayer consists of **hydrophilic (water-loving) heads** facing outward and **hydrophobic (water-hating) tails** facing inward.
- Lipid and protein molecules are joined together by **hydrogen bonds**.
- Thus, the arrangement is **protein-lipid-protein**, like a sandwich. Therefore, it is called the **Sandwich Model**.

B. Fluid Mosaic Model — Singer & Nicolson (1972)

- The membrane consists of a **phospholipid bilayer** with proteins embedded within it.
- Phospholipids are **amphipathic**, with **hydrophilic heads** and **hydrophobic tails**.
- Proteins are **asymmetrically arranged in a mosaic pattern**.
- Two main types of proteins:
 - **Peripheral (extrinsic) proteins** — lie on the membrane surface.
 - **Integral (intrinsic) proteins** — partially or completely embedded in the bilayer.
- Some integral proteins span the entire membrane and are called **transmembrane/channel proteins**.
- Some membrane proteins function in **transport and other cellular activities**.



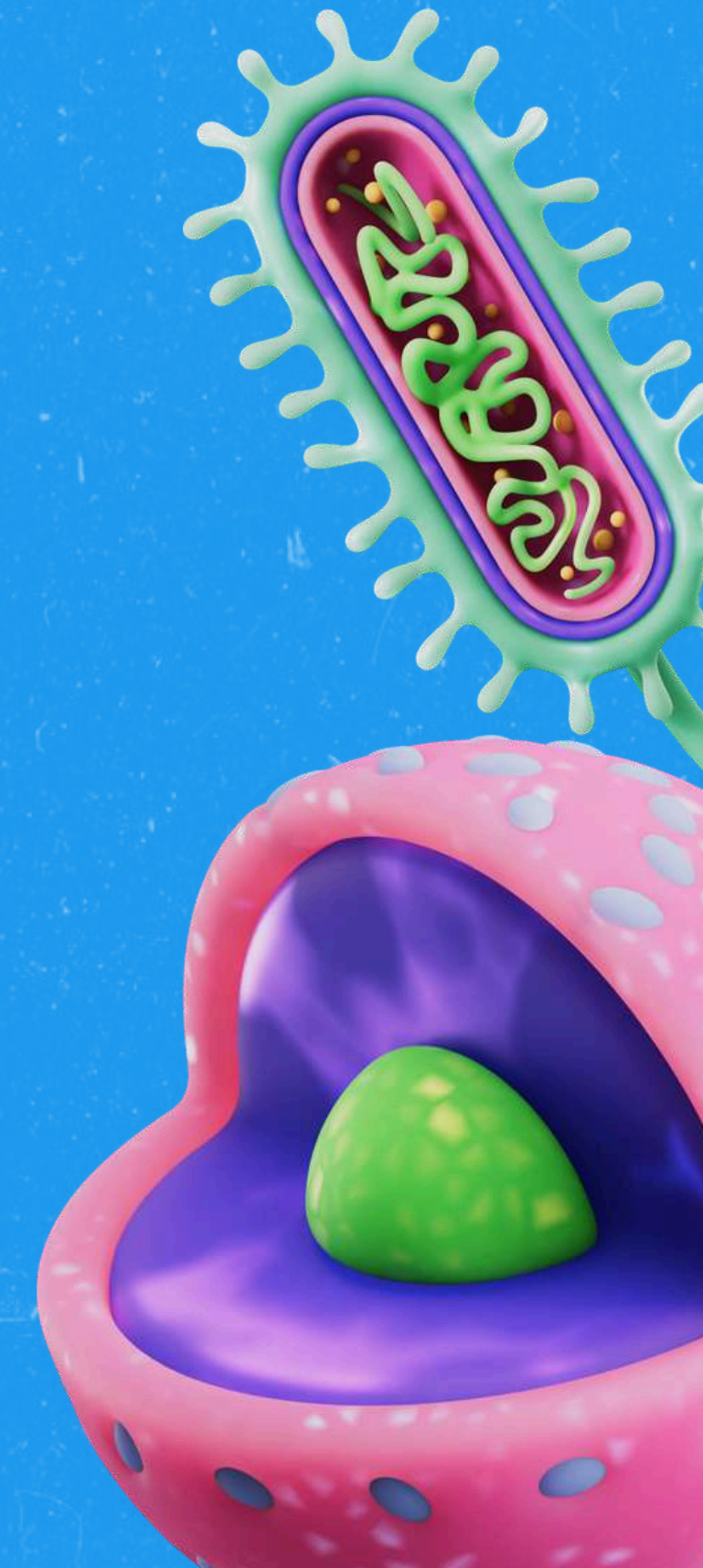
Cell Membrane - continued

C. Lipid Bilayer Model — Gorter & Grendel (1925)

- Proposed that the cell membrane is made of a **double layer of phospholipids**.
- **Hydrophilic heads** face the watery environments inside and outside the cell.
- **Hydrophobic tails** face inward toward each other.
- This arrangement forms a **selective barrier** around the cell.
- Their work provided early evidence for the **phospholipid bilayer**.

D. Unit Membrane Model — Robertson (1959)

- Proposed that all biological membranes have a **similar basic three-layered structure**.
- The membrane consists of a **lipid layer between two protein layers**.
- Under an electron microscope, it appears as **dark-light-dark bands**.
- This model suggested that the **cell membrane and organelle membranes share a common structure**.
- It later became limited because membranes are **more dynamic and complex** than a simple three-layer arrangement.

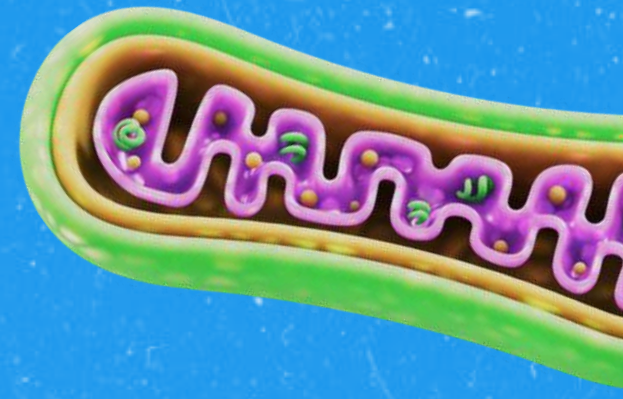


Functions of Cell Membrane

- **Main Function:** Regulates all molecular traffic entering and leaving the cell.
- **Diffusion:** Naturally moves gases, ions, and soluble substances from high to low concentrations.
- **Osmosis:** Specifically diffuses water or solvents across the semi-permeable membrane.
- **Active Transport:** Uses ATP energy to force molecules against their concentration gradient.
- **Carrier Proteins:** Relies on specific helper molecules called permeases and translocases.
- **Bulk Transport:** Moves large-scale packages via endocytosis (importing) and exocytosis (exporting).
- **Phagocytosis:** Performs "cell eating" by wrapping the membrane around large solid particles.



Cyanobacteria: Key Characteristics



Appearance

Cyanobacteria form rounded or filamentous colonies, often yellow-green, blue-green, or dark green.

Nature

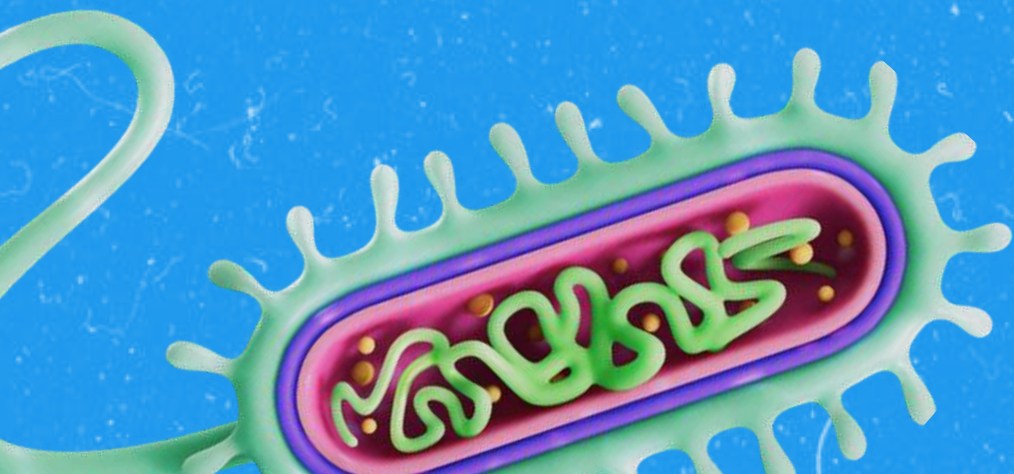
Commonly called blue-green algae, they are actually **photosynthetic bacteria** found in aquatic and moist environments.

Cellular Organization

They are **prokaryotic**, lacking a true nucleus and membrane-bound organelles.

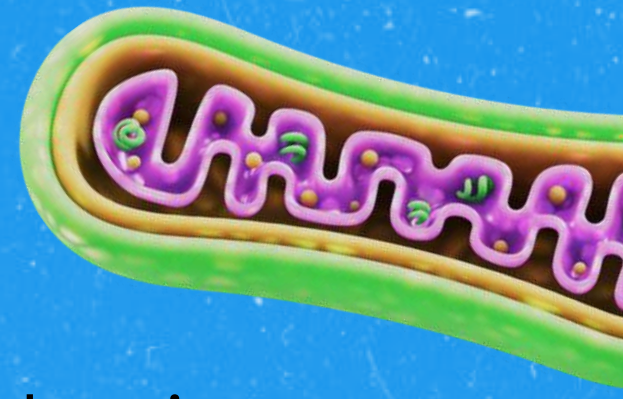
Importance

They perform **photosynthesis** and release oxygen; some can also **fix atmospheric nitrogen**.



Semi- autonomous organelles

ORGANELLES WITH THEIR OWN GENETIC MACHINERY—BUT NOT COMPLETE INDEPENDENCE

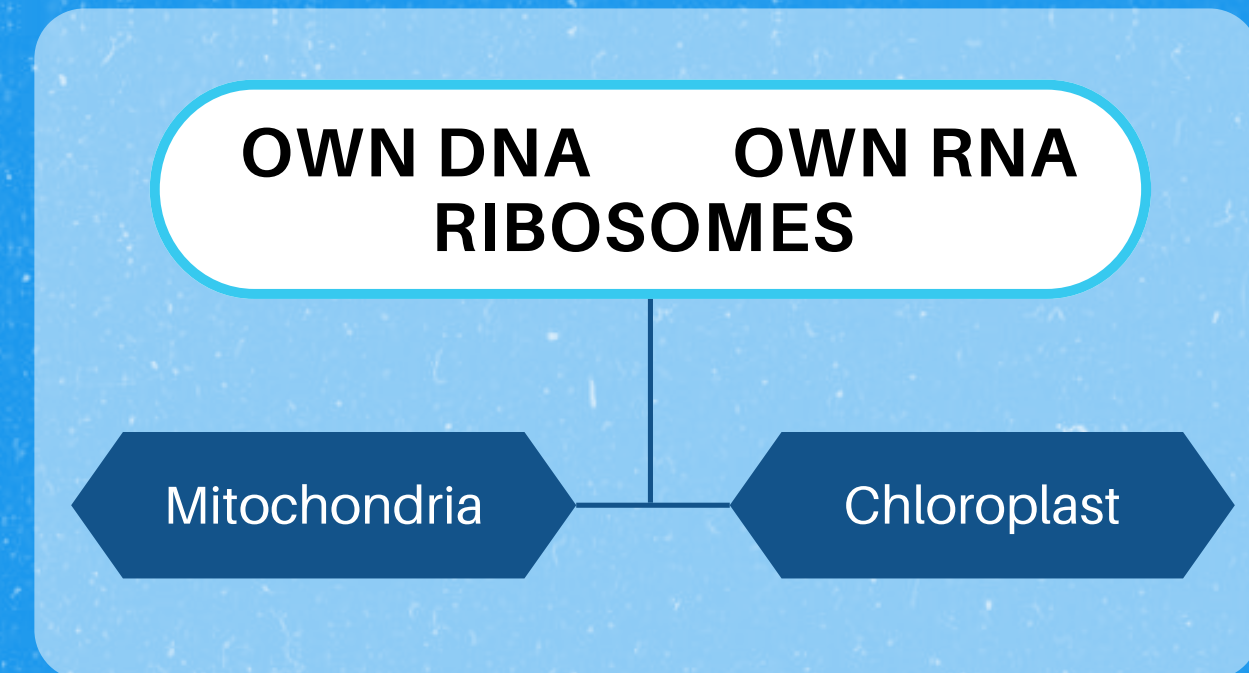


-Semi-autonomous organelles are organelles that possess some genetic and reproductive independence.

They contain:

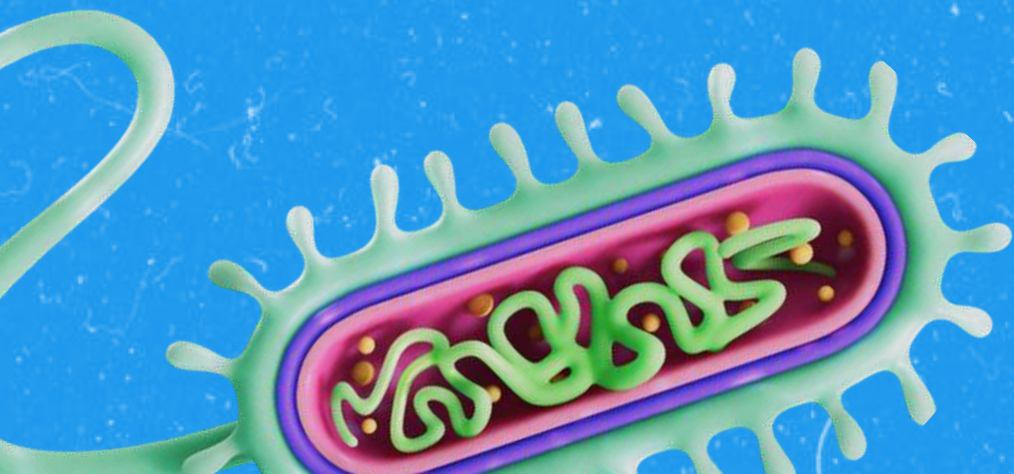
- Their own DNA
- RNA
- Ribosomes
- Machinery to synthesize some of their own proteins

-But not completely independent and still rely heavily on the nuclear genome and cytoplasm.



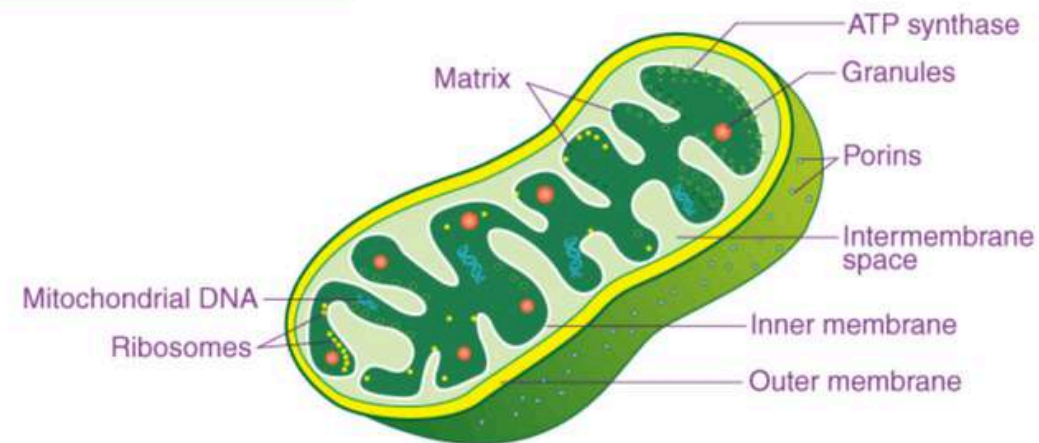
Major examples

- Mitochondria
- Chloroplasts — in plants and algae



Mitochondria

THE CLASSIC EXAMPLE OF A SEMI-AUTONOMOUS ORGANELLE



Mitochondria are double-membrane organelles found in most eukaryotic cells. They are major sites of **ATP production during aerobic respiration**, earning them the name “**powerhouses of the cell.**”

They possess:

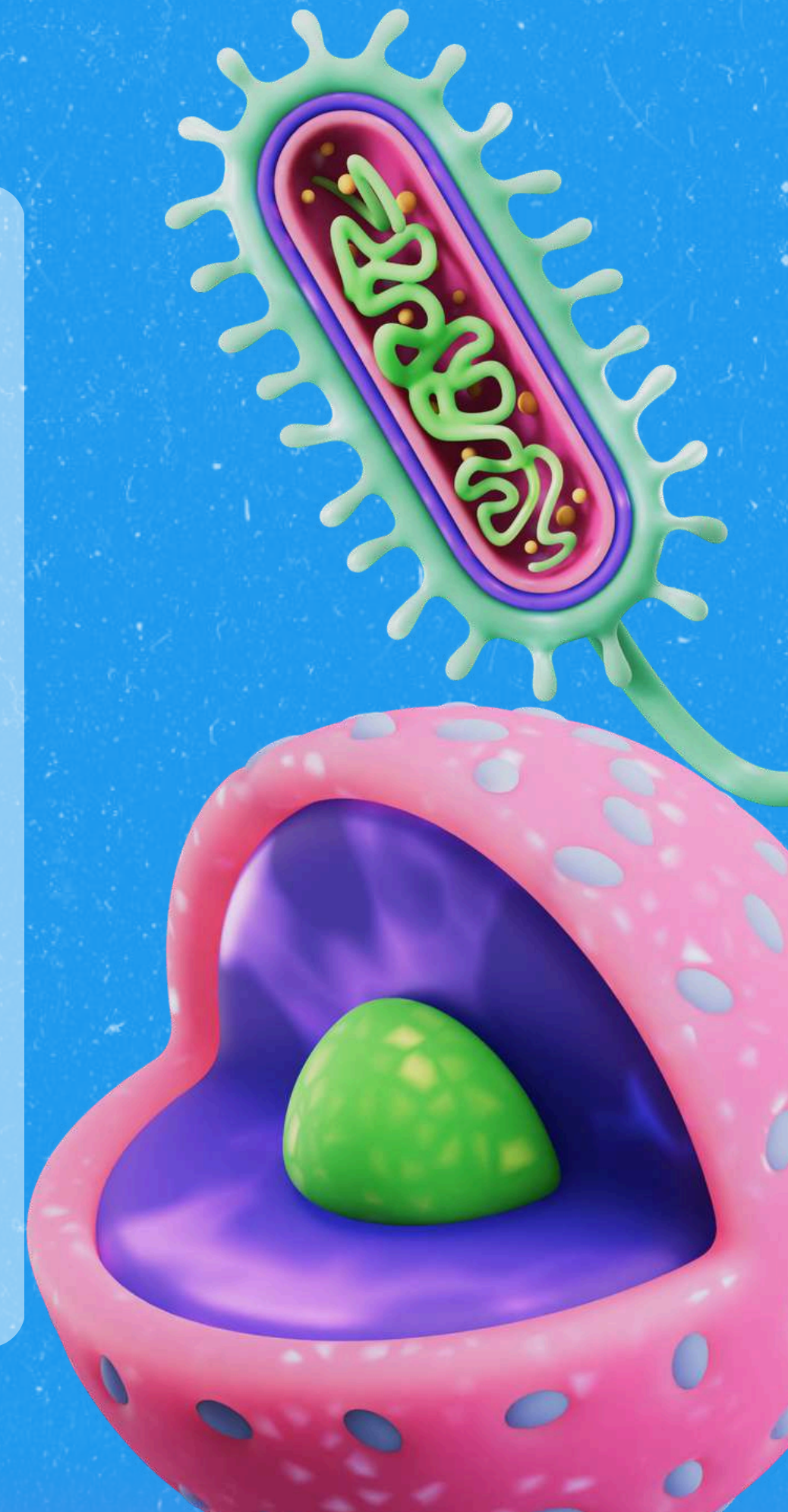
- **Circular mitochondrial DNA (mtDNA)**
- **RNA**
- **Prokaryote-like ribosomes**
- Machinery for synthesizing a **limited number of proteins**

However:

Most mitochondrial proteins are encoded by nuclear genes and imported into the mitochondrion

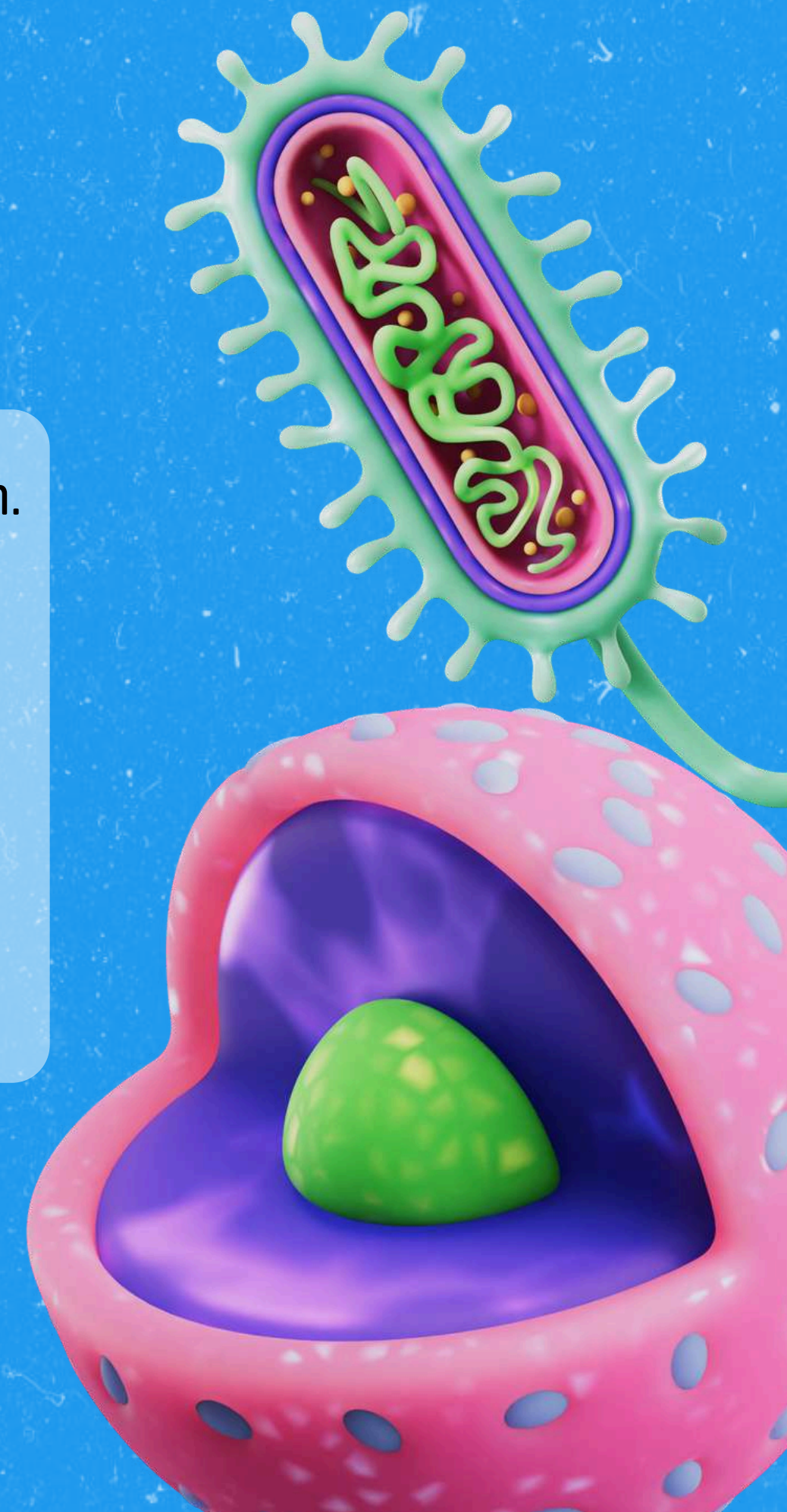
Structure & Anatomy of Mitochondria

- Mitochondria are usually rod-shaped or sausage shaped organelle
- Each mitochondrion has two membrane:
 - Outer membrane = smooth in structure
 - Inner membrane = folded inward to form many finger-like structure called cristae.
 - Cristae increases the surface area for cellular respiration
- Cristae contain many tennis racket like structure is called oxysomes or F^1 particle or elementary particle.
- The inner membrane enclose a fluid-filled substance called the matrix
- The matrix contain:
 - Circular DNA
 - RNA
 - 70s ribosomes
 - Protein and enzymes
 - Enzymes needed for the krebs cycle.
- The space between outer and inner membrane is called inner membrane space. or perimitochondrial space
- It contains enzymes/co-enzymes involved in processes such as fat oxidation



Functions of Mitochondria

- **ATP Production:** Generates energy through cellular respiration and oxidative phosphorylation.
- **Metabolic Pathways:** Houses the Krebs cycle and drives fatty acid beta-oxidation.
- **Cell Death Control:** Triggers apoptosis (programmed cell destruction) in damaged cells.
- **Calcium Homeostasis:** Stores and releases calcium ions to regulate cell signaling.
- **Heat Generation:** Produces body heat via proton leaks in non-shivering thermogenesis.
- **Heme Synthesis:** Initiates the biochemical production of hemoglobin for red blood cells.
- **Immune Signaling:** Activates antiviral signaling pathways to help fight cellular infections.
- **ROS Regulation:** Manages Reactive Oxygen Species to control cellular stress and signaling.





THANKYOU !

Any questions regarding our topic are
welcome.

